

OCR-RCNN: An Accurate and Efficient Framework for Elevator Button Recognition

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I. INTRODUCTION

Abstract—Autonomous elevator operation is considered a promising solution for mobile navigation in office buildings. As a fundamental function, elevator button recognition remains unsolved due to the challenging image conditions and severe data imbalance problem. In this article, we propose an accurate and efficient framework, named OCR-RCNN, for elevator button recognition. The framework is comprised of a region-based convolutional neural network (R-CNN)-based button detector and an attention-RNN-based character recognizer. Leveraging the two components, we further propose an end-to-end architecture and a cascaded architecture to explore the most effective network design for the framework. Moreover, a perspective distortion removal algorithm is also developed to enhance the inference performance of OCR-RCNN. Another key contribution of this work is that we release the first large-scale elevator panel dataset with 2005 images and 21 767 button labels. Extensive experiments are conducted on the released dataset and other two publicly available datasets. The proposed framework achieves an F1 score of 0.94, 1.00, and 1.00 in detection task, and an accuracy of 79.6% 96.5%, and 96.4% in character recognition task. The results demonstrate the advantages of our method, outperforming alternative strategies and other state-of-the-art methods in the literature. The data and code are available on the project webpage <https://github.com/zhudelong/ocr-rcnn-v2>.

Index Terms—Elevator button detection, elevator operation, optical character recognition, and service robot.

AUTONOMOUS elevator operation is considered a promising solution to navigate service robots freely inside of office buildings. The expansion of working space significantly increases the mobility of service robots and benefits human beings in more application scenarios [1], [2]. As an intelligent solution, autonomous elevator operation does not require human assistance or elevator retrofitting, thus bringing much convenience to robot users. In this work, we focus on one of the most fundamental problems in elevator operation, i.e., button recognition, which directly determines the performance of the entire operation system.

Button recognition refers to the task of *button detection* and *labeling* [3], which are both difficult problems due to the challenging image conditions, e.g., distortions, image blurs, and specular reflections. The large variety of panel designs and lack of low-level image features further increase the difficulties of button recognition. To tackle these problems, traditional methods [4]–[8] follow a two-stage processing pipeline, in which the buttons are first detected, and then an *optical character recognition* (OCR) algorithm [9] or a classification algorithm [5]–[8] is leveraged to label the detected buttons. Since these methods highly depend on hand-crafted features, their performances are not robust to environment changes, and the accuracy is also limited. In recent years, based on deep learning techniques, some multiobject detection frameworks are leveraged to conduct button recognition [10]–[13], in which the button detection and labeling tasks are simultaneously conducted by a detection network and a classification network, respectively. Although these methods achieve more accurate and robust performance in handling specific elevators, their generalization ability remains unverified due to the lack of large-scale elevator panel datasets.

In this work, we release the first large-scale elevator panel dataset and propose an OCR-RCNN framework to address the button recognition problem. Different from multiobject detection framework, OCR-RCNN leverages a specially designed OCR network, rather than the classification network, to conduct button labeling task. The idea is based on our observations of the dataset: 1) It is difficult to define all the button classes using a multiobject detection network, since there exist a large number of character combinations, e.g., *101*, *G1*, and once a new button type is added, the network has to be modified. On the contrary, OCR network is built upon a character dictionary and

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performs character-level recognition, e.g., *101* is sequentially labeled as (*1*, *0*, *1*), thus, can address this problem well. 2) The data imbalance problem is very severe in the dataset since low-rise buildings are more than high-rise ones, which makes multiobject detection models easily overfitted to classes with sufficient training samples [3]. By leveraging the attention-RNN, OCR network decomposes large floor numbers into independent characters, which are then recognized individually, thus, significantly reducing the number of categories to be recognized and easing the data imbalance problem.

Based on the OCR network, two button recognition architectures are designed: 1) The end-to-end architecture integrates an OCR network into the Faster RCNN [14], resulting in an end-to-end trainable network for simultaneous button detection and character recognition; 2) The cascaded architecture adopts a two-stage strategy, in which the buttons are first detected and then labeled by the OCR network. Both architectures take as input elevator panel images and output labeled bounding boxes for the recognized buttons. Based on the two architectures, several efficient implementations of OCR-RCNN are also presented to serve as baselines for the released dataset.

To further improve the inference performance of OCR-RCNN, we also propose a *perspective distortion removal* (PDR) algorithm. The algorithm first exploits the predicted bounding boxes to fit an optimal grid for the buttons, then takes the grid centers as target points to estimate distortion parameters. We formulate this process as the *maximum likelihood estimation* (MLE) problem of an underlying *Gaussian mixture model* (GMM), and an *expectation-maximization* (EM) based optimization algorithm is also derived to solve the MLE problem. After removing the distortion, the OCR-RCNN is then recalled to achieve a better button recognition result.

In the conference version [3], two contributions are claimed as follows.

- 1) We formalize the button labeling task as an OCR problem and implement an OCR network based on the additive attention-RNN.
- 2) Based on the OCR network and Faster R-CNN, we propose an end-to-end architecture for simultaneous button detection and character recognition.

The new contributions of this article are listed as follows.

- 1) We further extend the OCR network with an implementation of multiplicative attention-RNN and propose a new cascade architecture for OCR-RCNN framework.
- 2) We propose a PDR algorithm that can estimate distortion parameters with a single elevator panel image.
- 3) We release the first large-scale elevator panel dataset to benchmark elevator button recognition algorithms and further provide several OCR-RCNN implementations to serve as baselines.

The remainder of this article is organized as follows. Section II briefly surveys recent studies on elevator button recognition. Sections III and IV explain details of our proposed framework. Experimental results that validate the proposed method are presented and discussed in Section V. Finally, Section VI concludes this article.

II. RELATED WORK

Elevator button recognition is a typical industrial application of object detection and OCR techniques. Related studies can be divided into two categories according to different problem formalizations.

Multiobject detection based methods: Early studies on button recognition are typically based on a single elevator, and the problem is usually tackled by multiobject detection frameworks, in which the labeling task is conducted by a classification algorithm. Before the wide application of deep learning [15], [16], the performance of button recognition is very limited due to the lack of robust object detection methods, which is hard to meet the requirements of real-world applications. For instance, in the AAI Mobile Robot Challenge [17] and TechX Challenge [18], one of the missions is to navigate the robot inside a building by autonomously operating the elevators. Even with the elevator panel image provided beforehand, the difficulty of successfully operating the elevator is still outstanding. In one of the earliest studies, Kim *et al.* [6] constructed a button recognition system based on a template matching method, in which the template buttons need to be previously sampled from a target elevator. Since the matching process is based on hand-crafted features, this method is vulnerable to environment changes. Recently, Islam *et al.* [8] and Dong *et al.* [10] proposed an accurate recognition pipeline, in which the edge information of buttons is first leveraged to generate as many button candidates as possible, and then a deep classification neural network is developed to further label the candidates. The drawback of this method is that the candidate proposal process highly depends on parameter tuning, which causes much inconvenience in model deployment. In more recent studies, Yang *et al.* [12] utilized a multiobject detection neural network for simultaneous button detection and classification. Although only a limited number of button categories can be recognized, the robustness of this method is significantly improved. Similarly, Liu and Tian [11] also leverage an end-to-end neural network to conduct button recognition, but they additionally integrate a segmentation network to mask the operable areas, which is very useful in handling elevators with special button designs.

Character recognition based methods: By leveraging multiobject detection neural networks, the problem of button recognition for a specific elevator is well addressed by [11] and [12]. However, when handling previously unknown elevators, especially when there exist a lot of button categories to consider, the capability of these methods is very limited. Simply formulating the recognition task as a multiobject detection problem will limit the improvement of recognition performance due to the severe data imbalance problem, as analyzed in Section I. Another idea is, thus, to build the recognition system upon OCR techniques [19]. The first systematic work based on OCR is reported in [4], in which the authors use a sliding window method to conduct button detection. Based on the detection results, the text on a button is segmented into characters, which are then labeled individually by a light-weight classification network. Leveraging this pipeline, the authors report a recognition accuracy of 88.4% in a test set with 50 elevator panel images

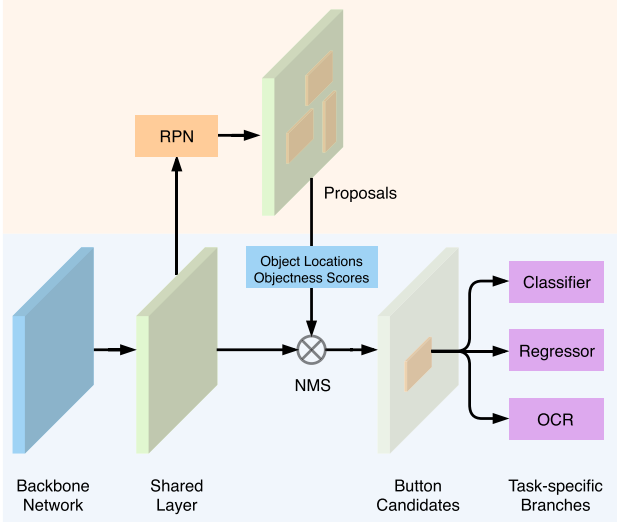


Fig. 2. End-to-end architecture. RPN and NMS are the abbreviation for *region proposal network* and *nonmaximum suppression*.

B. End-to-End OCR-RCNN

Architecture design: An overview of the end-to-end architecture is presented in Fig. 2, which is developed based on faster R-CNN [14]. The architecture is comprised of a *region proposal network* (RPN), the top part, and a *button recognition network* (BRN), the bottom part, which share the same feature layer extracted by the *backbone network*. When performing forward propagation, the RPN first generates a large number of button proposals with each of them being scored. Based on the scores and locations of the proposals, the *nonmaximum suppression* (NMS) is then conducted to remove the poor-quality proposals. The remaining ones are then leveraged to crop feature patches $\mathbf{f}$, i.e., the *button candidates*, from the shared layer. Finally, these candidates are passed to the *task-specific branches* for simultaneous button classification, location regression, and character recognition.

The advantages of this architecture lie in two aspects: 1) Multitask learning can improve the generalization ability of the model and make the learned features more discriminative. 2) The classification and OCR are independently conducted, and their predictions can be aggregated for ensemble learning to further improve the recognition performance. However, since the locations of the button candidates are not accurate, the accuracy of character recognition may be badly affected.

Loss function: The loss function is comprised of a classification loss L_c , a regression loss L_r , an OCR loss L_o , and an RPN loss L_p , weighted by different coefficients as follows:

$$L = \lambda_c L_c + \lambda_r L_r + \lambda_o L_o + \lambda_p L_p \quad (6)$$

where L_c is a typical *binary cross entropy* (BCE) loss

$$L_c = \frac{1}{N_c} \sum_{i=1}^{N_c} \text{BCE}(b_i, b_i^*), \quad b_i, b_i^* \in \{0, 1\}. \quad (7)$$

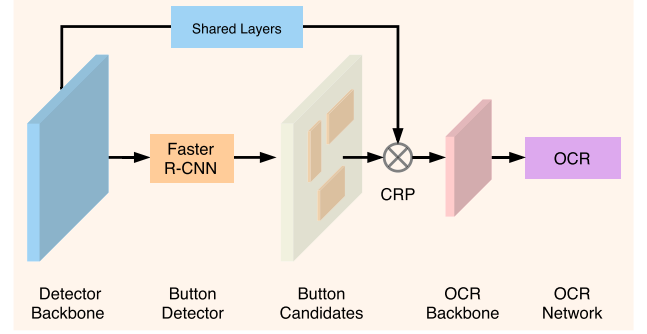


Fig. 3. Cascaded architecture. CRP represents *crop-and-resize*.

L_r is the Huber loss for robust regression [14]

$$L_r = \frac{1}{N_r} \sum_{i=1}^{N_r} \mathbb{1}[b_i \in \{1\}] \text{Huber}(\mathbf{p}_i, \mathbf{p}_i^*)$$

$$\text{Huber}(\mathbf{p}, \mathbf{p}^*) = \begin{cases} 0.5 \|\mathbf{p} - \mathbf{p}^*\|_2^2, & \|\mathbf{p} - \mathbf{p}^*\|_1 < 1 \\ \|\mathbf{p} - \mathbf{p}^*\|_1 - 0.5, & \text{otherwise} \end{cases}. \quad (8)$$

L_o is an instance-level *categorical cross entropy* (CCE) loss

$$L_o = \frac{1}{N_o} \sum_{i=1}^{N_o} \mathbb{1}[b_i \in \{1\}] \sum_{t=1}^T \text{CCE}(\mathbf{o}_i^t, \mathbf{c}_i^t). \quad (9)$$

It is worth noting that, N_r and N_o denote the number of positive button candidates. The RPN loss L_p is comprised of an objectness loss and a localization loss, which have the same formula with L_c and L_r , respectively. This is different from the typical multiobject detection problem, in which L_c is usually formulated as a CCE loss, which essentially has a different formula with the objectness loss (BCE).

C. Cascaded OCR-RCNN

Architecture design: An overview of the cascaded architecture is presented in Fig. 3. A major improvement is that the outputs of the Regressor, as shown in Fig. 2, are leveraged to generate button candidates, which are with higher regression accuracy than the RPN proposals in end-to-end architecture. In forward propagation, the detector first enumerates all the button candidates, based on which the corresponding feature patches are then cropped from specific layers of the detector backbone, e.g., the first layer, and resized to the same dimension. This process is similar to the button proposal step in RPN. Finally, the cropped patches are further processed by the OCR backbone, e.g., MobileNet, and the output $\mathbf{f}$ is then fed into the OCR network for character recognition.

The advantage of the cascaded design lies in the improvement of OCR accuracy, which benefits from the high-quality button candidates and the integration of OCR backbone. The drawback is that the button detector and OCR module have to be trained separately, and some extra engineering work is, thus, needed to cascade the trained models for further improving the two-stage inference efficiency.



Fig. 4. Original image (left) and the rectified image (right). The green diamonds indicate the centers of the detected buttons, and the red crosses represent the vertices of the grid.

Loss function: The loss function contains the same components with (6), but the network is first trained with a detection loss

$$L = \lambda_c L_c + \lambda_r L_r + \lambda_p L_p \quad (10)$$

and then the OCR loss L_o . The only difference is that b_i and p_i in (7)–(9) are now predicted by faster R-CNN rather than the RPN.

IV. PERSPECTIVE DISTORTION REMOVAL

In this section, we first formalize the problem with a GMM model, then derive the solution based on the EM framework.

Throughout this section, the following notations are used.

- 1) $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N] \in \mathbb{R}^{2 \times N}$ - the centers of detected buttons in image plane.
- 2) $\hat{\mathbf{X}} = [\hat{\mathbf{x}}_1, \dots, \hat{\mathbf{x}}_N] \in \mathbb{R}^{3 \times N}$ - the centers of detected buttons in normalized image plane.
- 3) $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_K] \in \mathbb{R}^{2 \times K}$ - the centroids of GMM model in image plane.
- 4) $\hat{\mathbf{U}} = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_K] \in \mathbb{R}^{3 \times K}$ - the centroids of GMM model in normalized image plane.
- 5) $\mathbf{z} \in \mathbb{R}^K$ - the hidden variable with a one-hot encoding scheme, which determines the correspondences between button centers and GMM centroids.
- 6) $\mathbf{o} = [o_x, o_y]^T$ - the origin of a two-dimensional (2-D) grid, i.e., the top-left vertex of the grid in Fig. 4.
- 7) $\mathbf{d} = [d_x, d_y]^T$ - the edge length between grid vertices in x and y directions in Fig. 4.
- 8) $\mathbf{s} = [i, j]^T$ - the 2-D coordinate of a grid vertex, which is indexed by k , $0 \leq k \leq K$.
- 9) $\Sigma = \text{diag}([\sigma_x, \sigma_y])$ - the diagonal covariance matrix shared by all GMM components, $\sigma = [\sigma_x, \sigma_y]^T$.
- 10) $\Theta = (\mathbf{o}, \mathbf{d}, \sigma)$ - the parameters of GMM to be estimated.
- 11) $\xi = [\theta, t]^T$ - the transformation (distortion parameters) between the current and desired angle of views, where θ is the angle-axis representation of rotation.
- 12) $\mathbf{R}(\theta)$ - the matrix representation of rotation.

A. Problem Formalization

The goal of perspective distortion removal is to estimate ξ by aligning two groups of feature points, i.e., the $\mathbf{X}$ and $\mathbf{U}$, in this work. This process is also called *point set registration* [24], which requires that both $\mathbf{X}$ and $\mathbf{U}$ are known. However, $\mathbf{U}$ is an unknown variable in our problem, thus, we have to estimate it first. Once $\mathbf{U}$ is obtained, ξ can, then, be calculated with typical registration methods. To estimate $\mathbf{U}$, we first define an initial grid $(\mathbf{o}, \mathbf{d})$ based on the detected buttons, shown in Fig. 4, and then $\mathbf{U}$ can be parameterized by $\mathbf{u}_k = \mathbf{o} + \mathbf{d} \cdot \mathbf{s}_k$. We then estimate the optimal $\mathbf{o}$ and $\mathbf{d}$ by solving the MLE problem of an underlying GMM.

This idea is first mentioned in [4] for refining the predicted bounding boxes, and it requires images without any distortion. In contrast, our method aims to remove perspective distortions, and thus, the problem formulation has an essential difference. Moreover, we derive a closed-form solution of Θ , based on which a complete EM framework is presented to estimate ξ .

To align $\mathbf{X}$ and $\mathbf{U}$, we follow the convention [24] and assume $\mathbf{X}$ is sampled from a GMM that takes $\mathbf{U}$ as centroids

$$p(\mathbf{x}|\Theta) = \sum_{\mathbf{z}} p(\mathbf{z})p(\mathbf{x}|\mathbf{z}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}|\mathbf{u}_k, \Sigma) \quad (11)$$

where $p(\mathbf{z}) \in \{\pi_1, \dots, \pi_K\}$ is a discrete distribution of $\mathbf{z}$, which is a prior that directs the mixture process of GMM. $p(\mathbf{z})$ is assumed to be an uniform distribution, i.e., $\pi_k = 1/K$.

To account for noises and outliers, another uniform distribution is added into the GMM

$$p(\mathbf{x}|\Theta) = \frac{(1-\alpha)}{C} + \alpha \sum_{k=1}^K \frac{1}{K} \mathcal{N}(\mathbf{x}|\mathbf{u}_k, \Sigma) \quad (12)$$

where α is the weight parameter. To make $p(\mathbf{x})$ a distribution in image plane, the following constraint should be satisfied:

$$\sum_{\mathbf{x}[1]=1}^w \sum_{\mathbf{x}[2]=1}^h p(\mathbf{x}|\Theta) = 1 \quad (13)$$

where the operator $[\cdot]$ is used to index the element of vector $\mathbf{x}$. The solution of (12) and (13) gives an approximation of C , i.e., $C \approx w \times h$, where w, h are the width and height of the image plane, respectively.

For simplicity, we add an additional dimension to $\mathbf{z}$, thus, $\mathbf{z} \in \mathbb{R}^{K+1}$, and use a more compact formula for (12)

$$p(\mathbf{x}|\Theta) = \sum_{k=1}^{K+1} \pi'_k p(\mathbf{x}|\mathbf{z}) \quad (14)$$

where $\pi'_k = \alpha \pi_k$ for $k = 1, \dots, K$ and $\pi'_{K+1} = 1 - \alpha$. Based on (11)–(14), the joint distribution of $\mathbf{x}$ and $\mathbf{z}$, $p(\mathbf{x}, \mathbf{z})$, can be derived as follows:

$$\begin{aligned} p(\mathbf{x}, \mathbf{z}_k = 1) &= \alpha \mathcal{N}(\mathbf{x}|\mathbf{u}_k, \Sigma)/K, \text{ for } k = 1, \dots, K \\ p(\mathbf{x}, \mathbf{z}_{K+1} = 1) &= (1 - \alpha)/C, \text{ for } k = K + 1. \end{aligned} \quad (15)$$

Given the predicted button centers $\mathbf{X}$, an optimal solution of Θ can be achieved by minimizing the following cost function:

$$E(\Theta) = - \sum_{n=1}^N \ln \sum_{k=1}^{K+1} \pi'_k p(\mathbf{x}|\mathbf{z}). \quad (16)$$

The solution Θ defines the best grid $\mathbf{U}$ for the detected buttons $\mathbf{X}$, and the best correspondence between $\mathbf{x}$ and $\mathbf{u}$ can also be obtained through the joint distribution $p(\mathbf{x}, \mathbf{z})$ (as we shall see). Denote a corresponded point pair by $(\mathbf{x}_n, \mathbf{u}_n)$, the distortion parameter ξ can be estimated by solving the following least-square problem in the normalized image plane:

$$\begin{aligned} \xi^* &= \arg \min_{\xi} \frac{1}{2} \sum_{n=1}^N \mathbf{c}_n^T \mathbf{c}_n \\ \mathbf{c}_n &= \hat{\mathbf{x}}_n - (\mathbf{R}(\theta) \hat{\mathbf{u}}_n + \mathbf{t}). \end{aligned} \quad (17)$$

B. Parameter Estimation

Equation (16) is also called *complete log likelihood* function, which is difficult to optimize directly, since the addition operation is inside the $\ln$ operator. A general tool to solve this problem is the EM framework [25], which instead optimizes the *expected log likelihood* function of (14). The EM framework contains two iteratively performed steps: E-step and M-step. In the E-step, we use the best estimation of Θ achieved so far, or an initial value at the beginning of the algorithm, to calculate the posterior

$$p(\mathbf{z}|\mathbf{x}) = \frac{p(\mathbf{x}, \mathbf{z})}{p(\mathbf{x})} \propto p(\mathbf{x}, \mathbf{z}) \quad (18)$$

which is essentially a discrete distribution table with each element recording a probability value $\gamma(n, k) = p(\mathbf{z}_k = 1|\mathbf{x}_n)$ that specifies the correspondence between $\mathbf{x}_n$ and $\mathbf{u}_k$.

In the M step, the posterior is leveraged to optimize the expected log likelihood and update the parameter Θ

$$\begin{aligned} \Theta^{\text{new}} &= \arg \max_{\Theta} Q(\Theta, \Theta^{\text{old}}) \\ &= \arg \max_{\Theta} \sum_{n=1}^N \sum_{\mathbf{z}} p(\mathbf{z}|\mathbf{x}_n, \Theta^{\text{old}}) \ln p(\mathbf{x}_n, \mathbf{z}|\Theta) \\ &= \arg \max_{\Theta} \sum_{n=1}^N \sum_{\mathbf{z}} p(\mathbf{z}|\mathbf{x}_n, \Theta^{\text{old}}) \ln p(\mathbf{x}_n|\mathbf{z}, \Theta) \\ &= \arg \min_{\Theta} \sum_{n=1}^N \sum_{k=1}^K \gamma_{nk} (\mathbf{e}_n^T \Sigma^{-1} \mathbf{e}_n + \ln |\Sigma|) \end{aligned} \quad (19)$$

where $\mathbf{e}_n = \mathbf{x}_n - (\mathbf{o} + \mathbf{d} \cdot \mathbf{s}_k)$, and γ_{nk} is an abbreviation for $\gamma(n, k)$. The derivative of Q w.r.t. Θ is as follows:

$$\frac{\partial Q}{\partial \Theta} = 2 \sum_{n,k} \gamma_{nk} (\Sigma^{-1} \mathbf{e}_n)^T \frac{\partial \mathbf{e}_n}{\partial \Theta} \quad (20)$$

$$\frac{\partial \mathbf{e}_n}{\partial \Theta} = \begin{bmatrix} -1 & 0 & -i_k & 0 \\ 0 & -1 & 0 & -j_k \end{bmatrix}. \quad (21)$$

By making $\frac{\partial Q}{\partial \Theta} = 0$ and assuming the covariance matrix Σ is diagonal, we have

$$\begin{aligned} \sum_{n,k} \gamma_{nk} \sigma \cdot (\mathbf{x}_n - (\mathbf{o} + \mathbf{d} \cdot \mathbf{s}_k)) &= \mathbf{0} \\ \sum_{n,k} \gamma_{nk} \sigma \cdot \mathbf{s}_k \cdot (\mathbf{x}_n - (\mathbf{o} + \mathbf{d} \cdot \mathbf{s}_k)) &= \mathbf{0}. \end{aligned} \quad (22)$$

Through solving the equations, a closed-form solution of Θ can be achieved

$$\begin{aligned} \mathbf{o} &= \frac{\sum_{n,k} \gamma_{nk} (\mathbf{x}_n - \mathbf{d} \cdot \mathbf{s}_k)}{\sum_{n,k} \gamma_{nk}} \\ \mathbf{d} &= \frac{\sum_{n,k} \gamma_{nk} \mathbf{s}_k (\mathbf{x}_n - \mathbf{o})}{\sum_{n,k} \gamma_{nk} (\mathbf{s}_k \cdot \mathbf{s}_k)} \\ \sigma &= \frac{1}{N} \sum_{n=1}^N \gamma_{nk} (\mathbf{e}_n \cdot \mathbf{e}_n). \end{aligned} \quad (23)$$

It is worth noted that the division and multiplication in (22) and (23) are performed elementwise. The E-step and M-step iteratively execute until convergence.

Based on the estimated Θ , $\mathbf{U}$ is obtained according to $\mathbf{u}_k = \mathbf{o} + \mathbf{d} \cdot \mathbf{s}_k$ and then used to estimate the distortion parameters in (17). Leveraging the estimated ξ , we are able to remove the perspective distortions by performing an inverse warping process. Note that the perspective distortion comes from the relative pose between the camera and elevator panel, and the distortion parameter ξ is essentially an estimation of this pose. Therefore, the PDR can also be leveraged to guide the motion of the camera to actively remove the distortion.

The advantages of the proposed PDR algorithm lie in two aspects: 1) The algorithm adopts the centers of the detected buttons as feature points and avoids the expensive detection and matching process of low-level image features, exhibiting more robust, and efficient performance; 2) The estimated grid contains important semantic information that can be further used to correct mislabeled buttons. There are also some drawbacks: 1) The algorithm is specially designed for processing elevator panel images, thus cannot be generalized to other applications; 2) The feature points are sparse in some cases and may fail the estimation process.

V. EXPERIMENT RESULTS

In this section, we first introduce a large-scale elevator panel dataset, then conduct some ablation studies to evaluate the performance of the OCR network and detection network. After that, we make an overall comparison between the proposed OCR-RCNN and state-of-the-art methods, and finally present some discussions about the results. All the experiments are conducted with a GTX TITAN X 12 GB GPU.

A. Elevator Panel Dataset

To evaluate our method, we release an elevator panel dataset with 2005 panel images and 21 767 button labels. Compared with existing datasets [4], [10] that have a limited number of buttons,

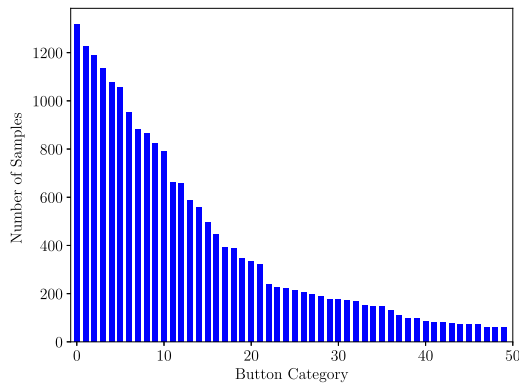


Fig. 5. Number of button samples in top-fifty categories.

TABLE I
DESIGNED CHARACTER DICTIONARY FOR THE DATASET

'0': 0	'1': 1	'2': 2	'3': 3	'4': 4	'5': 5
'6': 6	'7': 7	'8': 8	'9': 9	'A': 10	'B': 11
'C': 12	'D': 13	'E': 14	'F': 15	'G': 16	'H': 17
'I': 18	'J': 19	'K': 20	'L': 21	'M': 22	'N': 23
'O': 24	'P': 25	'R': 26	'S': 27	'T': 28	'U': 29
'V': 30	'X': 31	'Z': 32	'<': 33	'>': 34	'(': 35
')': 36	'\$': 37	'#': 38	'&': 39	's': 40	'-': 41
'*': 42	'%': 43	'?': 44	'!': 45	'φ': 46	

the released dataset is the first large-scale publicly available dataset in this research area. The dataset contains 365 classes of buttons with various shapes and appearances, covering the majority of existing button categories.

The number of button samples in top-fifty classes is shown in Fig. 5. As is indicated, there is a severe data imbalance problem in the dataset, and nearly 3/4 classes lack enough samples. This problem is caused by the fact that the number of low-rise buildings is always larger than that of high-rise ones. Any large-scale elevator panel dataset will inevitably encounter this problem. In this work, an OCR network is leveraged to address the data imbalance problem. The text on a button is first decomposed into a sequence of characters, e.g., $-L1$ becomes $(-, L, 1)$, which are then recognized individually. To facilitate this process, we build up a character dictionary for the dataset, as shown in Table I. Based on the dictionary, the OCR network can encode and recognize thousands of button categories without changing the network design, whereas typical classification models can only process a fixed number of button classes.

The dataset is randomly divided into a training set and a test set with a ratio of 7:3. During the training phase, some data augmentations are adopted, including random *rotation*, *shift*, *zoom*, *shear*, and *brightness*. In terms of the evaluation metrics, we use the average recall (AR), average precision (.), and F1 score for button detection, and average accuracy (ACC) for character recognition. The inference efficiency is also considered, as it is an important factor for robot control.

B. Ablation Studies

The goal of the first experiment is to verify the effectiveness of OCR network in addressing the data imbalance problem. To

achieve this goal, as presented in Tables II and III, we compare the button labeling performance of classification models adopted by multiobject detection frameworks, and the proposed OCR networks based on two efficient backbones.

- 1) *Classification + imbalance data*: This model is trained without any data balancing strategy and serves as the baseline. The initial learning rate is set to $1e-3$ and decays by $0.1x$ every 17 epochs, with a total of 50 training epochs, and the following models adopt the same settings.
- 2) *Classification + oversampled data*: The training data is uniformly sampled by classes regardless of the number of samples in a specific category, which aims to balance the data by oversampling classes without enough samples.
- 3) *Classification + weighted classes*: In contrast with the oversampling strategy, the data imbalance problem in this model is tackled by weighting the loss values according to the number of samples in each category.
- 4) *OCR + multiplicative attention*: This model is trained using the multiplicative attention, the first case in (4).
- 5) *OCR + additive attention*: This model is trained using the additive attention, the second case in (4). Both OCR models do not adopt any data balancing strategy.

As is indicated in Table II, both the OCR models significantly outperform the classification-based approaches. First, compared with the baseline (row 1), the recognition accuracy of OCR models is increased by about 3.3% (InceptionV3) and 2.5% (MobileNetV2) on the first dataset, and about 6.8% (InceptionV3) and 3.3% (MobileNetV2) on the second dataset, which demonstrates the advantages of the OCR models in conducting button labeling task. Second, compared with the typical data balancing strategies (row 2–3), the OCR models still maintain significant advantages and exhibit the capability of addressing data imbalance problem. Finally, it can be seen that the network scales of the OCR models are much smaller than those of the classification models. Hence, they are more suitable for embedded applications in the area of robotics. In terms of the inference time, it is slightly increased due to the expensive implementation of the RNN decoding process, which remains to be optimized.

To further verify the capability of OCR models in tackling data imbalance problem, we remove the data of top-21 classes from test set, i.e., 0–9, G, up, down, open, close, stop, alarm, call, empty, blur, and text, yielding a subset of 3930 buttons. Based on this subset, we conduct a comparison experiment and the results are presented in Table III. By comparing with the ACC-1 in Table II, it can be seen that the performance of the baseline model (row 1) is decreased, while the classification models with data balancing strategies (row 2–3) maintain consistent performance. The performance of OCR models, especially the additive model, is even improved, which indicates that the OCR models can effectively abstract character-level features and generalize well in dealing with button categories from high-rise buildings. The results again prove the effectiveness of OCR models in addressing the data imbalance problem.

In the second experiment, we evaluate the button detection performance of OCR-RCNN with different detector backbones, which aims to provide a benchmark for model selection in different application scenarios. We test three light-weight

TABLE II

PERFORMANCE OF CLASSIFICATION AND OCR MODELS. ACC-1, -2, AND -3 ARE THE AVERAGE ACCURACY OF CHARACTER RECOGNITION ON OUR RELEASED DATASET AND OTHER TWO DATASETS INTRODUCED BY [4] AND [10]. PARAMS AND TIME INDICATE THE PARAMETER SCALE (IN MILLIONS) AND INFERENCE TIME (IN SECONDS PER BATCH) OF THE MODEL, RESPECTIVELY

Method	InceptionV3					MobileNetV2				
	ACC-1	ACC-2	ACC-3	Params	Time (s)	ACC-1	ACC-2	ACC-3	Params	Time (s)
Classification + Imbalanced Data	0.8423	0.8666	0.9750	39.16	0.091	0.8511	0.8957	1.0000	44.74	0.054
Classification + Oversampled Data	0.8191	0.8673	0.9750	39.16	0.092	0.8333	0.8965	0.9500	44.74	0.055
Classification + Weighted Classes	0.8437	0.8702	0.9250	39.16	0.090	0.8554	0.9045	0.9875	44.74	0.053
OCR + Multiplicative Attention	0.8786	0.9500	0.9750	23.10	0.104	0.8737	0.9227	1.0000	3.13	0.066
OCR + Additive Attention	0.8744	0.9198	0.9500	22.69	0.108	0.8784	0.9344	1.0000	2.87	0.071

TABLE III

PERFORMANCE OF CLASSIFICATION AND OCR MODELS (ACC-1) ON A SUBSET OF BUTTONS IN HIGH-RISE BUILDINGS

Network	InceptionV3	MobileNetV2
Classification + Imbalanced Data	0.7959	0.7880
Classification + Oversampled Data	0.8198	0.8565
Classification + Weighted Classes	0.8326	0.8104
OCR + Multiplicative Attention	0.8893	0.8583
OCR + Additive Attention	0.8898	0.8827

TABLE IV

DETECTION PERFORMANCE OF OCR-RCNN WITH DIFFERENT BACKBONES ON THE RELEASED DATASET

Network	AR	AP ^{0.5IoU}	F1	Time (s)
MobileNetV1	0.860	0.995	0.923	0.149
InceptionV2	0.889	0.998	0.940	0.235
InceptionV3	0.899	0.997	0.945	0.291
ResNet50	0.898	0.997	0.945	0.297
InceptionResNetV2	0.915	0.999	0.955	0.342

backbones (row 1–3) and two high-capacity backbones (row 4–5), and the experiment results are presented in Table IV. As is indicated, the InceptionResNetV2 based implementation achieves the best detection performance, which has the largest network capacity. Benefiting from the compact network design, the MobileNetV1 [26] and InceptionV2/V3 [27] based implementations show significant advantages in inference efficiency. Meanwhile, these light-weight implementations also show competitive detection performance compared with the high-capacity ones, which indicates that the task of button detection does not require much learning capacity and can be effectively addressed by efficient backbones.

C. Overall Performance

We compare the overall performance of OCR-RCNN with existing studies in this section. For the end-to-end architecture, an InceptionResNetV2 is adopted as the backbone [3]. For the cascaded architecture, we use an efficient implementation, in which an InceptionV2 and a MobileNetV2 are adopted as the detector and OCR backbones, respectively. In both architectures, the additive attention based OCR is implemented.

The first dataset used for evaluation is introduced by [4], which contains 686 test buttons. All the images are captured

in favorable conditions and without any distortion. A character dictionary with 15 characters, i.e., 1–9, G, L, B, open, close, and alarm, is designed for the dataset. The categories that can be recognized are number buttons (combinations of 1–9), three letter buttons (G, L, B), and three function buttons (open, close, alarm). The evaluation results on this dataset are presented in row 1–3 of Table V.

The second dataset is introduced by [10], which contains 4 elevator panel images with 60 button labels. No character dictionary is designed for this dataset. Instead, the authors adopt the idea of multi-object detection for button recognition. The buttons are divided into 15 classes, i.e., 0–9, open, close, alarm, others, and negatives. The evaluation results are presented in row 4–6 of Table V.

The third one is our released dataset, which has a larger testing scale and more challenging image conditions, as indicated by column 4–6. The associated dictionary contains 47 characters, which is capable of encoding thousands of button categories. The evaluation results on this dataset are presented in row 7, 9, and 11 in Table V. Herein, since the cascaded OCR-RCNN adopts InceptionV2 as detection backbone, we additionally test an InceptionResNetV2-based implementation in row 9–10 to further evaluate the cascaded OCR-RCNN.

As indicated in Table V, the proposed method significantly outperforms the state-of-the-art methods. Especially, on the first two datasets, almost all the buttons are correctly detected and labeled by the OCR-RCNN models. On our released dataset, the detection task remains high performance, while the accuracy of character recognition is not very satisfying. As emphasized in previous sections, this is caused by the challenging image conditions and the severe data imbalance problem in the dataset. By comparing the end-to-end architecture (row 7) with the cascaded one (row 9 and 11), we can see that the cascaded OCR-RCNN presents a higher OCR accuracy and better inference efficiency. The improvements mainly benefit from the more accurate button proposals and more efficient backbones. We visualize some detection and recognition results of the cascaded OCR-RCNN in Fig. 6, to demonstrate the ability of the proposed method in handling different types of elevator panels.

To verify the effectiveness of the PDR algorithm, we rectify the images in the dataset first, and then conduct button recognition with OCR-RCNN. The experiment results are presented in row 8, 10, and 12 of Table V. Through comparing row 7, 9, 11 with row 8, 10, 12 accordingly, we can see that both the detection and OCR performance of OCR-RCNN are further

TABLE V

PERFORMANCE OF DIFFERENT BUTTON RECOGNITION METHODS. THE AR, AP, AND ACC ARE CALCULATED USING THE METHOD IN [4], [10]. THE "IR" MEANS THAT AN INCEPTIONRESNETV2 IS ADOPTED AS THE DETECTION BACKBONE IN THE CASCADED OCR-RCNN. THE "R" INDICATES THAT THE IMAGE DISTORTIONS IN THE DATASET ARE RECTIFIED. THE SIGN "-" REPRESENTS *NOT REPORTED*

No.	Model	Testing Scale			Detection			OCR	Efficiency
		Buttons	Characters	Distortion	AR	AP ^{0.5IoU}	F1	ACC	Time (s)
1	Klingbeil's work [4]	686	15	N	0.862	0.881	0.875	0.884	-
2	End-to-end OCR-RCNN [3]	686	15	N	1.000	1.000	1.000	0.817	1.196
3	Cascaded OCR-RCNN	686	15	N	0.990	1.000	0.995	0.965	0.614
4	Dong's work [10]	60	-	Y	0.900	0.876	0.888	-	-
5	End-to-end OCR-RCNN [3]	60	-	Y	1.000	1.000	1.000	0.850	2.268
6	Cascaded OCR-RCNN	60	-	Y	1.000	1.000	1.000	0.964	0.457
7	End-to-end OCR-RCNN [3]	6869	47	Y	0.9204	0.9978	0.9575	0.6203	0.8073
8	End-to-end OCR-RCNN [3]	6869	47	R	0.9235	0.9978	0.9592	0.6229	0.8149
9	Cascaded OCR-RCNN (IR)	6869	47	Y	0.9150	0.9986	0.9549	0.7863	0.4678
10	Cascaded OCR-RCNN (IR)	6869	47	R	0.9192	0.9986	0.9572	0.7918	0.5137
11	Cascaded OCR-RCNN	6869	47	Y	0.8892	0.9977	0.9403	0.7921	0.3591
12	Cascaded OCR-RCNN	6869	47	R	0.8948	0.9979	0.9436	0.7964	0.3888



Fig. 6. Demonstration of button recognition results. The recognized characters are visualized at the center of detected buttons. The green text on the bounding box represents the confidence of button detection. If the label and operable region share the same button entity (the first row), they are simply regarded as a single button instance. When the label and operable region are side by side (the second row), 1) if the two regions are located in different entities, we take them as two independent buttons; 2) otherwise (the first image in the second row), we regard them as a single button. To identify operable regions for robotic manipulation, we may apply contour detections [10] or integrate a segmentation branch [11] to mask the corresponding areas in the predicted bounding boxes.

improved, which demonstrates the advantages of applying the PDR algorithm. It is noteworthy that the improvement here is not very significant, which is due to the fact that the models are already robust to certain degrees of perspective distortions after training with various data augmentations. Hence, for some distorted images, the character recognition shows the same results before and after applying the PDR algorithm.

VI. CONCLUSION

In this work, we released the first large-scale elevator panel dataset and propose an OCR-RCNN framework for addressing the button recognition problem. We designed two fundamental architectures for the framework and provide several efficient implementations, including the additive and multiplicative attention based OCR modules, different button detection modules, and the PDR algorithm. Intensive evaluations demonstrate the high performance of the OCR-RCNN framework. Along with the dataset, several network implementations are also released to benchmark future methods and facilitate related studies on autonomous elevator operation.

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